Narrative Quantification 2.0

A Framework for the Observability of Reasoning Dynamics in Structured State Spaces

Gemmina Intelligence LLC | acta-aiie.org | Version 2.0.1

A Framework for the Observability of Reasoning Dynamics in Structured State Spaces

Abstract

We present *Narrative Quantification 2.0*, a formal framework that redefines AI reasoning as a measurable dynamical system over structured state representations. Rather than treating language generation as an opaque autoregressive process, we introduce a tripartite state architecture in which each narrative state is decomposed into an invariant core (s*core), a dynamic reasoning field (*sfield), and a projection state (s_projection). This separation enforces non-circular dependencies and enables explicit observability of internal reasoning dynamics. To ensure state identifiability under representational equivalence, we define a deterministic cryptographic mapping (T25) over the invariant core using canonicalized serialization. This guarantees reproducibility and exact state matching across reasoning trajectories. We further introduce a structure-preserving projection phi: \mathcal{S -> R^3, mapping high-dimensional narrative states into an interpretable semantic space defined by Directionality, Intensity, and Stability. The projection is trained via metric learning to approximate structural divergence Delta V, enabling geometric analysis of narrative relationships under bounded distortion. Reasoning is formalized as a discrete-time dynamical system governed by entropy-modulated exploration and convergence. Candidate states are evaluated using an objective function and selected via Boltzmann sampling with temperature T(H), ensuring controlled exploration under high entropy and stabilization as entropy decreases. Empirically, we validate the framework using a dataset of 1,563 events (9,378 narrative instances). Temporal analysis reveals that the delay between structurally related narrative pairs follows an exponential-like distribution, and is statistically inconsistent with randomized baselines (p < 10^-170), indicating a structured temporal response. Additionally, entropy comparisons show that only pairs sharing an identical invariant core exhibit significant entropy reduction ($p < 0.001$), supporting the presence of structural coupling rather than random aggregation. These results demonstrate that narrative evolution exhibits measurable dynamical structure, characterized by temporal response patterns, entropy reduction, and geometric consistency. The proposed framework provides a foundation for transforming AI reasoning from an unobservable generative process into a structured, verifiable, and analyzable system.

1. Introduction

Narrative analysis plays a central role in understanding large-scale information systems, including news dissemination, social media dynamics, and knowledge propagation. Despite its importance, existing approaches often lack formal structure and reproducibility, making it difficult to verify results or compare outcomes across systems.

Most current methods rely on probabilistic language models, embedding-based similarity, or heuristic feature extraction. While these approaches are effective for generating or clustering text, they typically do not provide a **deterministic and verifiable representation** of narrative states. As a result, identical inputs may produce different outputs across runs or systems, limiting reproducibility and interpretability.

This limitation becomes critical in scenarios where consistent tracking and comparison of narratives are required, such as longitudinal analysis, cross-platform monitoring, and system-level auditing. Without a canonical representation, it is difficult to determine whether two narratives correspond to the same underlying event or to quantify how narratives evolve over time.

To address these challenges, we propose a **state-space framework for narrative quantification**. In this framework, narratives are represented as deterministic vectors derived from observable attributes of events, enabling consistent construction across independent systems. This representation allows narrative evolution to be analyzed as transitions between states in a measurable space.

Building on this formulation, we introduce:

- **Narrative Change Distance (NCD)**, a metric for quantifying changes between consecutive narrative states,
- **entropy-based measures** for characterizing the distribution of narrative states, and
- a deterministic hashing mechanism, **state_hash**, which provides a canonical identifier for narrative states and enables reproducible tracking.

Unlike existing approaches that emphasize probabilistic generation or semantic similarity, our framework focuses on **measurement, consistency, and verification**. This shift allows narrative analysis to be treated as a structured and reproducible process rather than an approximate or heuristic one.

We evaluate the proposed framework on a dataset of 1,563 events comprising 9,378 narrative instances. The results demonstrate that narrative states can be constructed deterministically, compared consistently using the proposed distance metric, and reproduced across independent executions using state_hash.

The primary contribution of this work is the establishment of a **verifiable foundation for narrative analysis**, enabling consistent representation, comparison, and tracking of narrative dynamics across systems.

---

2. The AIIE Protocol and Tripartite State Architecture

2.1 The Need for Structural Separation

INarrative data is inherently heterogeneous, combining invariant event attributes with context-dependent and representation-specific features. In many existing approaches, these elements are treated as a single undifferentiated representation, which can lead to ambiguity, instability, and reduced comparability across instances.

For example, descriptions of the same underlying event may vary significantly depending on contextual framing, linguistic variation, or surrounding information. Without structural separation, these variations can obscure the underlying identity of the event and complicate the analysis of narrative evolution.

To address this issue, it is necessary to distinguish between:

- attributes that remain stable across representations,
- attributes that depend on contextual embedding, and
- attributes that are directly used for observable analysis.

This separation enables more precise control over how narrative states are constructed and compared.

Formal Motivation

Let a narrative representation be constructed from input data xxx. A nave formulation maps this directly to a single vector:

xSx \rightarrow SxS

However, such a representation conflates multiple sources of variation, making it difficult to:

- identify invariant properties of the underlying event,
- isolate context-dependent effects,
- ensure consistency across different observations.

Structured Alternative

We instead introduce a structured formulation:

S=(score,sfield,sprojection)S = (s{core}, s{field}, s_{projection})S=(score,sfield,sprojection)

where each component serves a distinct functional role.

This decomposition allows narrative states to be constructed in a way that preserves invariant information while explicitly accounting for contextual variation and observable representation.

Design Rationale

The separation into three components is motivated by the need to satisfy the following properties:

- **Consistency**: identical events should produce identical core representations
- **Stability**: contextual variation should not distort invariant attributes
- **Comparability**: states should be comparable across different contexts
- **Reproducibility**: independent systems should reconstruct equivalent states

By enforcing this structure, the framework avoids conflating fundamentally different types of information within a single representation.

Implication

This structural separation is a prerequisite for:

- defining meaningful distance metrics,
- constructing reproducible identifiers (state_hash),
- and analyzing narrative dynamics over time.

2.2 Tripartite State Formulation

We formalize the narrative state as a tripartite vector composed of three distinct components:

S=(score,sfield,sprojection)S = (s{core}, s{field}, s_{projection})S=(score,sfield,sprojection)

where each component belongs to a finite-dimensional real-valued space:

scoreRnc,sfieldRnf,sprojectionRnps{core} \in \mathbb{R}^{nc}, \quad s{field} \in \mathbb{R}^{nf}, \quad s{projection} \in \mathbb{R}^{np}scoreRnc,sfieldRnf,sprojectionRnp

and the total state is defined as:

SRn,n=nc+nf+npS \in \mathbb{R}^{n}, \quad n = nc + nf + n_pSRn,n=nc+nf+np

---

2.2.1 Core State scores_{core}score

The core state represents invariant attributes of the underlying event. It is constructed from features that remain stable across different representations of the same event.

Formally, let xxx denote input data associated with an event. The core state is defined as:

score=fcore(x)s{core} = f{core}(x)score=fcore(x)

where fcoref_{core}fcore is a deterministic mapping that extracts invariant features.

Properties

- **Determinism**: identical inputs yield identical core states
- **Invariance**: robust to contextual or representational variation
- **Uniqueness (empirical)**: distinct events produce distinct core states

---

Examples of Features

- entity identifiers
- temporal markers
- structural descriptors

2.2.2 Field State sfields_{field}sfield

The field state captures contextual attributes associated with the event. These attributes may vary depending on surrounding information but are systematically derived.

sfield=ffield(x,C)s{field} = f{field}(x, \mathcal{C})sfield=ffield(x,C)

where C\mathcal{C}C denotes contextual data.

Properties

- **Context-dependent**: varies with surrounding information
- **Deterministic**: identical inputs and context produce identical outputs
- **Extensible**: additional contextual features can be incorporated

---

Examples of Features

- co-occurring entities
- relational signals
- distributional patterns

2.2.3 Projection State sprojections_{projection}sprojection

The projection state defines the observable representation used for downstream analysis.

sprojection=fproj(score,sfield)s{projection} = f{proj}(s{core}, s{field})sprojection=fproj(score,sfield)

Properties

- **Derived**: constructed from core and field states
- **Normalized**: ensures comparability across instances
- **Task-compatible**: supports distance measurement and analysis

---

Typical Forms

- normalized embeddings
- reduced-dimensional representations
- feature transformations

2.2.4 Deterministic Composition

The full state is constructed via a deterministic pipeline:

S=f(x,C)=(fcore(x),ffield(x,C),fproj(score,sfield))S = f(x, \mathcal{C}) = (f*{core}(x), f*{field}(x, \mathcal{C}), f*{proj}(s*{core}, s_{field}))S=f(x,C)=(fcore(x),ffield(x,C),fproj(score,sfield))

This ensures:

- reproducibility across systems
- consistency across executions
- compatibility with deterministic identifiers (state_hash)

2.2.5 Normalization Constraint

To ensure comparability across states, we enforce:

S=1\| S \| = 1S=1

This normalization guarantees that distance-based comparisons are invariant to scale.

2.2.6 Design Guarantees

The tripartite formulation satisfies the following guarantees:

- **Separation**: invariant and contextual information are explicitly distinguished
- **Consistency**: identical inputs map to identical states
- **Comparability**: distances between states are meaningful
- **Reproducibility**: independent systems reconstruct equivalent states

2.3 Identifiability

This functional decoupling guarantees strict identifiability. We guarantee the existence of an extraction function \exists f: S -> s*core that reliably isolates the invariant properties from the full state. We assert that two reasoning processes have reached identical cognitive conclusions if and only if their s*core components match up to a canonical equivalence class, regardless of divergence in s*field (computational path) or s*projection (rendering format):

$s*core^A \sim s*core^B \implies \text{T25(A) = \text{T25(B)$

3. s_core and the Cryptographically Verifiable Invariant State (T25)

3.1 Defining the Invariant Core

To ensure consistency and avoid circular dependencies in state construction, we define the core state scores_{core}score exclusively from invariant observations and stabilized components:

score={Event,sstable}={T0121,(T2635)}s*{core} = \{ \text{Event}, s*{stable} \} = \{T*{01\dots21}, \Pi(T*{26\dots35})\}score={Event,sstable}={T0121,(T2635)}

This formulation ensures that the core state depends only on canonicalized observational inputs and stabilized features.

Importantly, the semantic position T22T22T22 is not included in scores_{core}score, but is instead defined as a deterministic function of invariant inputs:

T22=(T0121)T22 = \phi(T_{01\dots21})T22=(T0121)

This eliminates circular dependencies by ensuring that derived representations are functions of primary observations, rather than components of the core state itself.

3.2 The Stability Operator \Pi

While dynamic variables T2635T_{26\dots35}T2635 may vary over time, certain aspects of these variables exhibit stable behavior. To extract these components, we define a stability operator \Pi as a mapping to a discrete space:

:Rk{0,1}m\Pi: \mathbb{R}^k \rightarrow \{0,1\}^m:Rk{0,1}m

The operator is implemented as an indicator function over the dynamic variables:

(T2635)=I[f(T2635)>]\Pi(T{26\dots35}) = \mathbb{I}[f(T{26\dots35}) > \tau](T2635)=I[f(T2635)>]

where \tau is a predefined threshold.

Properties

- **Deterministic**: identical inputs produce identical outputs
- **Idempotent**: repeated application does not change the result
- **Robust**: stable under bounded temporal variation

This operation extracts discrete indicators from continuous dynamics, allowing stable components to be incorporated into the core state while excluding transient variation.

3.3 T25: The Deterministic State Hash

To ensure reproducibility and consistency across systems, we define T25 as a deterministic mapping from the core state space to a fixed-length binary representation:

T25:Score{0,1}256\text{T25}: \mathcal{S}_{core} \rightarrow \{0,1\}^{256}T25:Score{0,1}256

T25 is constructed in two steps:

- **Canonicalization**:

The core state scores_{core}score is serialized using a deterministic function C()\mathcal{C}(\cdot)C() compliant with the JSON Canonicalization Scheme (RFC 8785).

- **Hashing**:

A cryptographic hash function HHH (SHA-256) is applied:

T25=H(C(score))\text{T25} = H(\mathcal{C}(s_{core}))T25=H(C(score))

Properties

- **Deterministic**: identical core states yield identical hashes
- **Representation-invariant**: independent of serialization order
- **Collision-resistant (empirical)**

This construction ensures that T25 functions as a canonical identifier for narrative states across independent systems.

3.4 Methodological Clarification

T25 is used exclusively as an internal identifier for:

- state consistency
- deduplication
- exact state comparison

It does not assume or require any form of distributed consensus, trust model, or external validation mechanism.

---

4. Structure-Preserving Projection (phi: R^d -> R^3)

4.1 Objective of Semantic Embedding

We define the structural divergence between two narrative states as the Euclidean distance between their vector representations:

Vij=v(si)v(sj)2\Delta V{ij} = \| \mathbf{v}(si) - \mathbf{v}(sj) \|2Vij=v(si)v(sj)2

where v(s)Rd\mathbf{v}(s) \in \mathbb{R}^dv(s)Rd is a deterministic feature vector derived from the state sss. This vector is constructed from normalized components of scores_{core}score and selected stable features.

Clarification

The mapping v:SRd\mathbf{v}: \mathcal{S} \rightarrow \mathbb{R}^dv:SRd is fully deterministic and shared across systems, ensuring consistent distance computation.

Narrative states exist in a high-dimensional space Rd\mathbb{R}^dRd, where d3d \gg 3d3. While this representation captures structural complexity, it is not directly suitable for visualization or low-dimensional reasoning.

To address this, we introduce a projection:

:SR3\phi: \mathcal{S} \rightarrow \mathbb{R}^3:SR3

that maps narrative states into a low-dimensional coordinate system.

Requirement

The projection is designed to approximately preserve structural divergence:

(si)(sj)2Vij\|\phi(si) - \phi(sj)\|2 \approx \Delta V{ij}(si)(sj)2Vij

This defines a task-specific embedding rather than a generic dimensionality reduction.

4.2 Formulation of the Semantic Axes

To ensure interpretability, we construct three axes using deterministic feature extractors derived from the state components.

Axis 1: Directionality (xxx)

Measures directional relationships within the state representation:

x=1ZxeEwedir(e)x = \frac{1}{Zx} \sum{e \in E} w_e \cdot \text{dir}(e)x=Zx1eEwedir(e)

where:

- EEE: set of directed relations extracted from the state
- dir(e){1,+1}\text{dir}(e) \in \{-1, +1\}dir(e){1,+1}
- wew_ewe: weighting coefficient

Axis 2: Intensity (yyy)

Measures the magnitude of underlying activity signals:

y=1ZyaAmay = \frac{1}{Zy} \sum{a \in A} \| \mathbf{m}_a \|y=Zy1aAma

where:

- AAA: set of entities
- ma\mathbf{m}_ama: feature vector associated with entity aaa

---

Axis 3: Stability (zzz)

Measures concentration of the state distribution:

z=1H(sfield)Hmaxz = 1 - \frac{H(s{field})}{H{\max}}z=1HmaxH(sfield)

where:

- H()H(\cdot)H(): entropy defined in Section 4
- HmaxH_{\max}Hmax: maximum entropy

4.3 Projection Optimization

The base feature extractors provide interpretability but do not guarantee distance preservation. Therefore, we define the optimal projection as:

=argminEi,j[((si)(sj)2Vij)2]\phi^* = \arg\min{\phi} \mathbb{E}{i,j} \left[(\|\phi(si) - \phi(sj)\|2 - \Delta V{ij})^2 \right]=argminEi,j[((si)(sj)2Vij)2]

This objective ensures that distances in the projected space approximate structural divergence.

4.4 Hybrid Projection Formulation

We define the projection as a combination of components derived from the state:

(s)=P0(score)+Q(sfield)\phi(s) = P \phi0(s{core}) + Q \psi(s_{field})(s)=P0(score)+Q(sfield)

where:

- 0:ScoreR3\phi0: \mathcal{S}{core} \rightarrow \mathbb{R}^30:ScoreR3
- :SfieldRk\psi: \mathcal{S}_{field} \rightarrow \mathbb{R}^k:SfieldRk
- PR33,QR3kP \in \mathbb{R}^{3 \times 3}, Q \in \mathbb{R}^{3 \times k}PR33,QR3k

---

Initialization

Matrices PPP and QQQ are initialized using Principal Component Analysis (PCA) and refined through optimization of the projection objective.

4.5 Metric Distortion Bounds

Since exact isometric embedding into R3\mathbb{R}^3R3 is generally not possible, we evaluate the projection using bi-Lipschitz bounds:

c1Vij(si)(sj)2c2Vijc*1 \Delta V*{ij} \leq \|\phi(s*i) - \phi(s*j)\|*2 \leq c*2 \Delta V_{ij}c1Vij(si)(sj)2c2Vij

where c1,c2>0c*1, c*2 > 0c1,c2>0 are empirically estimated.

Interpretation

These bounds ensure that relative distances are preserved within controlled distortion, providing a stable basis for downstream analysis.

5. Entropy and Adaptive State Selection

5.1 State Transition as a Dynamical Process

We model the reasoning process as a discrete-time dynamical system over the state space:

st+1=F(st,It)s*{t+1} = \mathcal{F}(s*t, I_t)st+1=F(st,It)

where:

- stSs_t \in \mathcal{S}stS: current state
- ItI_tIt: input at time ttt
- F\mathcal{F}F: deterministic transition operator

Interpretation

The transition operator updates the state by integrating new information and recomputing derived components.

The dynamic component sfields*{field}sfield captures intermediate variations during this process, while score*s{core}score represents the stabilized outcome after integration.

5.2 Entropy Dynamics with Exploration Bound

We define entropy H(st)H(s_t)H(st) over the state distribution (Section 4).

A well-behaved system reduces entropy as evidence accumulates. However, temporary increases are allowed to account for exploration.

We model this as:

E[H(st+1)]H(st)+t\mathbb{E}[H(s*{t+1})] \leq H(s*t) + \epsilon_tE[H(st+1)]H(st)+t

where t0\epsilon_t \geq 0t0 represents bounded exploration.

Convergence Condition

To ensure stability, we require:

tt<\sum*t \epsilon*t < \inftytt<

Interpretation

- Early stages: higher entropy (broader exploration)
- Later stages: entropy decreases (stabilization)

5.3 Entropy-Adaptive Objective Function

We define an objective function with entropy-dependent weights:

J(st,It)=kk(H)fk(st,It)J(st, It) = \sum{k} \lambdak(H) \cdot fk(st, I_t)J(st,It)=kk(H)fk(st,It)

where H=H(sfield)H = H(s_{field})H=H(sfield).

Weight Definition

k(H)={k(1+H2)k{1,2}kmax(,1H2)k{3,4}\lambdak(H) = \begin{cases} \alphak (1 + H^2) & k \in \{1,2\} \\ \alpha_k \cdot \max(\eta, 1 - H^2) & k \in \{3,4\} \end{cases}k(H)={k(1+H2)kmax(,1H2)k{1,2}k{3,4}

Interpretation

- High entropy exploration-oriented terms dominate
- Low entropy stability-oriented terms dominate

Stability Constraint

A lower bound \eta ensures that stability-related terms remain active throughout the process.

5.4 Energy-Based Selection Mechanism

We define a probabilistic selection over candidate states:

P(s(i))=exp(Ji/T(H))jexp(Jj/T(H))P(s^{(i)}) = \frac{\exp(-\bar{J}i / T(H))}{\sumj \exp(-\bar{J}_j / T(H))}P(s(i))=jexp(Jj/T(H))exp(Ji/T(H))

Temperature Mapping

T(H)=Tmin+(1Tmin)HT(H) = T{min} + (1 - T{min}) HT(H)=Tmin+(1Tmin)H

where Tmin>0T_{min} > 0Tmin>0.

Score Normalization

Ji=Jimin(J)max(J)min(J)+\bar{J}i = \frac{Ji - \min(J)}{\max(J) - \min(J) + \epsilon}Ji=max(J)min(J)+Jimin(J)

Degenerate Case Handling

If score variance is below a threshold, a uniform distribution is used.

Interpretation

- High entropy broader sampling
- Low entropy concentrated selection

5.5 State Update and Consistency

The selected state is used to update:

- scores_{core}score (invariant component)

- sfields_{field}sfield (dynamic component)

The updated core state is then mapped to a deterministic identifier via T25.

---

6. Consistency Scoring and Structural Integrity

6.1 Continuous Consistency Penalty

To ensure smooth integration into the state transition process, we define a continuous consistency score:

C(s)=exp(kpkak(s))C(s) = \exp\left(- \sumk pk \cdot a_k(s) \right)C(s)=exp(kpkak(s))

where:

- ak(s)[0,1]a_k(s) \in [0,1]ak(s)[0,1]: magnitude of the kkk-th anomaly
- pkp_kpk: predefined penalty weight

Interpretation

The score C(s)(0,1]C(s) \in (0,1]C(s)(0,1] represents the degree of structural consistency of a state.

- C(s)1C(s) \approx 1C(s)1: highly consistent
- C(s)0C(s) \rightarrow 0C(s)0: inconsistent

Design Rationale

This formulation enables:

- continuous penalty application
- differentiability for optimization
- integration into probabilistic selection

Integration into Selection

The score is incorporated as a multiplicative factor:

P(s(i))C(s(i))exp(Ji/T(H))P'(s^{(i)}) \propto C(s^{(i)}) \cdot \exp(-\bar{J}_i / T(H))P(s(i))C(s(i))exp(Ji/T(H))

This reduces the probability of inconsistent states without enforcing hard constraints.

6.2 Geometric Interpretation in Projection Space

In the projected space R3\mathbb{R}^3R3, consistency influences the stability dimension.

Low-consistency states exhibit increased dispersion in the dynamic component, which can be interpreted as higher uncertainty in their spatial representation.

Operational Effect

- High C(s)C(s)C(s): concentrated representation
- Low C(s)C(s)C(s): dispersed representation

This behavior prevents inconsistent states from appearing artificially stable in downstream analysis.

6.3 Impact-Based State Transition

We define a normalized impact measure to quantify structural changes between states:

Impactnorm=Hamming(sbase,scandidate)Baseline\text{Impact}{norm} = \frac{\text{Hamming}(s{base}, s_{candidate})}{\text{Baseline}}Impactnorm=BaselineHamming(sbase,scandidate)

Thresholding

A transition is classified as a **significant update** when:

Impactnorm>impact\text{Impact}{norm} > \tau{impact}Impactnorm>impact

where impact\tau_{impact}impact is a predefined threshold.

Interpretation

This mechanism distinguishes:

- incremental updates (small impact)
- structural changes (large impact)

6.4 Role in the Framework

The consistency score and impact measure together provide:

- robustness against anomalous inputs
- controlled state updates
- improved stability of the transition process

7. State Re-expansion Dynamics

7.1 Motivation

During the state transition process, we observe cases where a previously stabilized state exhibits a sudden increase in variability following new input.

These events cannot be explained by gradual updates alone and require a formal mechanism to capture abrupt structural changes.

7.2 Formal Definition of Re-expansion Events

Let t2\sigma_t^2t2 denote the variance of selected state attributes at time ttt.

We define a **re-expansion event** at time ttt as:

REtt12high\text{RE}t \iff \sigma^2{t-1} < \theta{\text{low}} \;\wedge\; \sigma^2t > \theta_{\text{high}}REtt12high

where:

- low\theta_{\text{low}}low: threshold for stabilized states
- high\theta_{\text{high}}high: threshold for high variability

---

Confirmation Condition

To ensure robustness, the event must persist:

REt+1confirmedREtt+12>high\text{RE}{t+1}^{\text{confirmed}} \iff \text{RE}t \;\wedge\; \sigma^2{t+1} > \theta{\text{high}}REt+1confirmedREtt+12>high

---

Interpretation

This condition captures a transition from a low-variance (stable) regime to a high-variance (exploratory) regime.

7.3 Objective Function Perspective

Let J(s)J(s)J(s) denote the objective function defined in Section 5.

A re-expansion event corresponds to a temporary increase:

J(st+1)J(st)>J(s*{t+1}) - J(*st) > \deltaJ(st+1)J(st)>

where >0\delta > 0>0.

Interpretation

This indicates that the system departs from a previously stable configuration and re-enters a broader search region.

7.4 Statistical Validation

To distinguish meaningful re-expansion from random variation, we define a null model:

- independent sampling of state dimensions
- fixed entropy level

Hypothesis Test

A re-expansion is considered significant if:

- the displacement st+1st2\|s*{t+1}* - *s*t\|_2st+1st2
- and the variance increase

exceed the null expectation with statistical significance:

p<0.05p < 0.05p<0.05

7.5 Role in the Framework

Re-expansion dynamics enable the system to:

- recover from premature convergence
- adapt to new input
- maintain responsiveness under changing conditions

8.Temporal Response Function of Narrative Emergence

8.1 Definition of Temporal Response

To quantify the temporal dynamics of narrative emergence, we define the response delay between structurally related narrative states.

Let (s+, *s*-) denote a pair of narrative states associated with a shared invariant core s*core, where sigma(s*+) = +1 and sigma(s_-) = -1 represent opposing semantic orientations.

The temporal response delay Delta t is defined as:

t=tt+,t0\Delta t = t- - *t*+, \quad \Delta t \geq 0t=tt+,t0

where t+ *and t*- denote the timestamps of the primary and counter-narrative observations, respectively.

We consider only pairs satisfying:

T25(s+)=T25(s)\text{T25}(s+*) = \text{T25}(s*-)T25(s+)=T25(s)

ensuring that both states originate from an identical invariant core.

8.2 Empirical Distribution and Model Fitting

Using the collected dataset of 1,563 events (9,378 narrative instances), we extract all valid (s+, *s*-) pairs and compute the empirical distribution of Delta t.

The observed distribution is evaluated against a canonical exponential model:

p(t)=etp(\Delta t) = \lambda e^{-\lambda \Delta t}p(t)=et

where lambda is estimated via maximum likelihood:

^=1E[t]\hat{\lambda} = \frac{1}{\mathbb{E}[\Delta t]}^=E[t]1

Empirically, we obtain:

- Mean delay: \mathbb{E[Delta t] \approx 1.92 hours
- Estimated rate: \hat{lambda \approx 0.52

The exponential model provides a close fit to the observed decay pattern, indicating that the probability of counter-narrative emergence decreases over time.

8.3 Causal Structure Verification via Shuffled Baseline

To test whether the observed temporal structure arises from causal dependency rather than random coincidence, we construct a shuffled baseline.

Specifically, we randomly permute timestamps across narrative states while preserving the pairing structure, generating a null distribution where temporal order is destroyed.

We then perform a KolmogorovSmirnov (KS) test between the observed Delta t distribution and the shuffled baseline.

Result:

- KS statistic: D = 0.4917
- p-value: p < 10^-170

This result strongly rejects the null hypothesis that the observed temporal structure is consistent with random ordering.

8.4 Interpretation as a Response Function

The exponential decay behavior suggests that narrative emergence follows a response-like mechanism.

We define the **temporal response function** as:

R(t)=etR(t) = \lambda e^{-\lambda t}R(t)=et

which characterizes the likelihood of counter-narrative generation at time t after the initial observation.

This formulation does not assume strict determinism; rather, it indicates that the emergence process is statistically constrained by prior states.

Importantly, the rejection of the shuffled baseline implies that:

- narrative emergence is temporally structured, and
- opposing narratives are not independently sampled events.

8.5 Hazard Function Analysis

To further characterize the temporal dynamics of narrative emergence, we estimate the hazard function lambda(t) associated with the response delay distribution.

The hazard function is defined as:

(t)=p(t)1F(t)\lambda(t) = \frac{p(t)}{1 - F(t)}(t)=1F(t)p(t)

where p(t) denotes the empirical probability density of Delta t, and F(t) is the corresponding cumulative distribution function.

In practice, lambda(t) is estimated using a discretized histogram-based approximation of p(t) combined with the empirical survival function 1 - F(t).

Under a memoryless (Poisson) process, the hazard function remains constant over time. However, the empirical estimate exhibits systematic deviations from a constant rate, indicating that the emergence of counter-narratives is not governed by a purely memoryless process.

The inset shows the empirically estimated hazard function lambda(t) (see 8.5), which exhibits systematic deviations from a constant rate, suggesting departures from a memoryless (Poisson) response process.

8.6 Summary

The temporal analysis establishes three key results:

- The delay between opposing narratives follows an exponential-like decay distribution.
- The temporal structure is statistically incompatible with randomized baselines (p < 10^-170).
- The hazard function exhibits non-constant behavior, indicating structured, non-Poissonian dynamics.

Together, these findings support the interpretation that narrative emergence operates as a **response-driven process within a structured dynamical system**, rather than as independent stochastic events.

---

9. Controlled Validation Experiments

To complement the empirical observations in Section 9, we conduct controlled experiments to isolate structural effects under known conditions.

9.1 Temporal Structure Validation (Shuffled Baseline Test)

Method

We compare:

- observed time differences t\Delta tt
- time differences under randomized timestamp shuffling

using the KolmogorovSmirnov test.

Result

- KS statistic: D=0.49D = 0.49D=0.49
- p<10170p < 10^{-170}p<10170

Interpretation

The observed temporal structure significantly deviates from the shuffled baseline.

This indicates that the ordering of narrative instances is not random and exhibits structured temporal dependencies.

9.2 Structural Consistency Test (Permutation Test)

We analyze pairs of states associated with the same event.

Method

We compare divergence measures between:

- pairs sharing the same scores_{core}score
- randomly paired states

using permutation testing.

Result

- p<0.001p < 0.001p<0.001
- large effect size

Interpretation

The reduction in divergence is specific to structurally related pairs and cannot be explained by random pairing.

9.3 Predictive Consistency (Time-Split Validation)

Method

We perform time-based splitting:

- training set: 80%
- test set: 20%

and evaluate prediction of unseen state attributes.

Result

- predictive accuracy: ~0.8
- stable similarity metrics

Interpretation

The results suggest partial predictability in state evolution under the proposed framework.

9.4 Role of Simulation

The above experiments are conducted using a controlled simulation environment that reproduces the statistical properties observed in the empirical dataset.

This allows:

- isolation of causal factors
- validation under known assumptions
- reproducibility across environments

10. Conclusion

In this paper, we presented a structured framework for representing and analyzing narrative dynamics as state transitions in a measurable space. By shifting from token-level representations to state-based modeling, the proposed approach enables reasoning processes to be examined through observable quantities such as entropy, geometric distance, and state transitions.

We introduced a tripartite state architecture consisting of an invariant core scores*{core}score, a dynamic component sfields*{field}sfield, and a projection component sprojections_{projection}sprojection. This separation allows invariant attributes to be distinguished from contextual variation, enabling consistent state construction and comparison across systems.

To address the problem of representational variability, we defined a deterministic identifier (T25) based on canonicalization and cryptographic hashing. This mechanism ensures that identical states can be reliably identified and retrieved, supporting reproducibility and longitudinal analysis.

We further proposed a structure-preserving projection into a low-dimensional space and demonstrated that the resulting representation maintains meaningful relationships between states under controlled distortion. In addition, we modeled state transitions as an entropy-regulated process with adaptive selection, allowing both stabilization and re-expansion behaviors to be captured within a unified framework.

Experimental evaluation on a dataset of 1,563 events and 9,378 narrative instances showed that the proposed framework enables:

- deterministic state construction,
- consistent distance-based comparison,
- statistically significant structural patterns in temporal and entropy dynamics.

These results provide empirical support for treating narrative analysis as a structured and reproducible process.

Overall, the proposed framework establishes a foundation for observable and verifiable state-based analysis of narrative data. This enables consistent tracking, comparison, and evaluation of narrative dynamics, and provides a basis for future extensions in large-scale information systems and AI-based analysis.

---

11. Limitations and Broader Impacts

11.1 Limitations of the Projection Approximation

The structure-preserving projection \phi is evaluated using empirically estimated bi-Lipschitz bounds c1,c2c*1, c*2c1,c2 derived from the dataset D\mathcal{D}D.

As a result, performance may degrade in out-of-distribution scenarios where the underlying data differs significantly from the training distribution.

Future work includes adaptive projection mechanisms that adjust to varying entropy levels and data regimes.

11.2 Limitations of Deterministic Consistency Scoring

The consistency score relies on predefined anomaly functions and fixed penalty weights pkp_kpk.

While this design ensures transparency and reproducibility, it may not fully capture context-dependent inconsistencies in complex or ambiguous cases.

In practice, domain-specific calibration of anomaly functions and weights may be required.

11.3 Computational Overhead

The proposed framework introduces additional computational steps, including:

- deterministic state construction
- canonicalization of scores_{core}score
- probabilistic selection based on entropy

These steps increase latency compared to standard single-pass generation.

However, this trade-off is acceptable in applications where reproducibility, traceability, and consistency are prioritized.

11.4 Broader Impacts

The framework provides a structured approach for analyzing and comparing narrative states, which may contribute to improved transparency in AI-assisted analysis.

By enabling explicit representation and comparison of state transitions, the approach can help reduce reliance on opaque intermediate processes and support more interpretable system behavior.

In addition, the inclusion of diversity-aware retrieval mechanisms may help mitigate over-concentration in local regions of the state space, improving robustness under varying input conditions.

Appendix A: Algorithmic Implementation

A.1 Pseudocode of the Dynamical System

To ensure reproducibility across environments, the following pseudocode defines the deterministic state transition process.

All stochastic operations are executed under fixed seeds derived from the state identifier.

Narrative State Transition Loop (Deterministic Implementation)

import hashlib

def run*reasoning*cycle(event*data, human*intent, prev_state):

1. Deterministic Seed Initialization

current*seed = int(prev*state.state_hash[:8], 16)

set*random*seed(current_seed)

2. Entropy Measurement

```
H = measureentropyratio(prev_state)
# 3. Dynamic Weight Modulation
h_sq = H ** 2
weights = {
"regret": ALPHA1 * (1 + hsq),
"divergence": ALPHA2 * (1 + hsq),
"coherence": ALPHA3 * max(0.1, 1 - hsq),
"value": ALPHA4 * max(0.1, 1 - hsq)
}
# 4. Candidate Generation
N = 1 + floor(4 * prevstate.rupturerisk)
worldlines = generateworldlines(prevstate, event_data, count=N)
# 5. Scoring
for w in worldlines:
w.score = calculateobjectivej(w, weights, human_intent)
normalizedscores = minmax_normalize([w.score for w in worldlines])
# 6. Probabilistic Selection
T = 0.1 + 0.9 * H
probabilities = [exp(-s / T) for s in normalized_scores]
selectedindex = stochasticsample(probabilities, seed=current_seed)
# 7. State Finalization
nextstate = worldlines[selectedindex]
nextstate.statehash = generatejcssha256(nextstate.coredata)
return next_state
```

Reproducibility Guarantee

The system ensures reproducibility through:

- deterministic state construction
- canonical serialization (RFC 8785)
- fixed seed propagation via state_hash
- controlled stochastic sampling

Appendix B: Dataset and Training Procedures

B.1 Dataset Description

We utilize a dataset consisting of:

- **1,563 events**
- **9,378 narrative instances**

In addition, an extended dataset is under construction for future evaluation.

Extended Dataset (Ongoing Work)

- target size: 10,000 pairs
- annotated structural divergence V\Delta VV
- anomaly score vectors aka_kak

B.2 Annotation Protocol

Ground-truth labels are obtained through a human annotation process.

Setup

- 5 expert annotators
- predefined evaluation rubric

Evaluation Criteria

- causal direction changes
- variation in actor intent magnitude
- degree of state stabilization

Quality Control

- inter-annotator agreement measured via correlation
- disagreement cases resolved through consensus

B.3 Training of Projection Matrices

We train the projection matrices PPP and QQQ to approximate structural divergence.

Objective

L(P,Q)=Ei,j[((si)(sj)2Vij)2]+(PF2+QF2)\mathcal{L}(P, Q) = \mathbb{E}*{i,j} \left[(\|\phi(*si) - \phi(s*j)\|2 - \Delta V*{ij})^2 \right] + \lambda (\|P\|F^2 + \|Q\|_F^2)L(P,Q)=Ei,j[((si)(sj)2Vij)2]+(PF2+QF2)

Training Setup

- optimizer: AdamW
- learning rate: 31043 \times 10^{-4}3104
- batch size: 256
- epochs: 50
- early stopping: validation-based

Initialization

- P=I3P = I_3P=I3
- QU(0.01,0.01)Q \sim \mathcal{U}(-0.01, 0.01)QU(0.01,0.01)

Reproducibility

- fixed random seed: 42
- deterministic batching
- consistent validation split (20%)

Appendix C: Standardized Computational Specifications

C.1 T25 Hash Construction

The T25 identifier is computed as:

- Canonicalization using JSON Canonicalization Scheme (RFC 8785)
- SHA-256 hashing

T25=H(C(score))\text{T25} = H(\mathcal{C}(s_{core}))T25=H(C(score))

C.2 Structural Divergence (V\Delta VV)

- Feature extraction from scores_{core}score
- normalization
- Euclidean distance:

Vij=v(si)v(sj)2\Delta V*{ij}* = \|*mathbf{v}(s*i) - \mathbf{v}(s*j)*\|2Vij=v(si)v(sj)2

C.3 Temperature Function

T(H)=0.1+0.9HT(H) = 0.1 + 0.9HT(H)=0.1+0.9H

Rationale

- avoids zero-temperature collapse
- ensures bounded stochasticity

C.4 Boltzmann Selection

Pi=exp(Ji/T(H))jexp(Jj/T(H))P*i* = *\frac{\exp(-\bar{J}*i / T(H))}{\sum*j* *\exp(-\bar{J}*j / T(H))}Pi=jexp(Jj/T(H))exp(Ji/T(H))

Figure 1: Tripartite State Architecture

A narrative state s is formally structured into three functionally distinct components: the invariant core s*core, the dynamic reasoning field s*field, and the projection state s_projection.

Both s*core and s*field are computed in parallel from the input x, where s*core* = *f*core(x) captures invariant observational facts, and s*field* = *f*field(x, C) represents the dynamic reasoning process conditioned on contextual memory C.

The projection state is defined as s*projection* = *f*proj(s*core,* sfield), integrating invariant and dynamic components into a low-dimensional semantic representation.

The invariant core is subsequently canonicalized and mapped to a deterministic identifier T25 via a cryptographic hash function.

This architecture enforces a strict separation between invariant and dynamic components, preventing circular dependencies and ensuring that all derived representations are functions of canonicalized observations.

```
                  x (Input)
                 /        \
                ↓          ↓
         s_core = f_core(x)   s_field = f_field(x, C)
                \          /
                 ↓        ↓
              s_projection = f_proj(s_core, s_field)
                          ↓
                    T25(s_core)
```

Figure 2: State Transition Dynamics

At each time step, multiple candidate worldlines are generated and evaluated using the objective function J. The selection of the next state is performed via Boltzmann sampling, where the probability of each candidate is determined by a temperature parameter T(H) derived from the current entropy H.

This formulation enforces a strict separation between deterministic scoring and stochastic selection. High entropy leads to elevated temperature and promotes exploratory sampling, while decreasing entropy lowers the temperature, resulting in convergence toward low-energy states.

```
s_t
 ↓
[ Candidate Worldlines ]
 ↓
[ Scoring J ]
 ↓
[ Boltzmann Selection ]
 ↓
s_{t+1}
```

Figure 3: Temporal response distribution of counter-narrative emergence.

The histogram shows the empirical distribution of response delays Delta t = t- - *t*+ for narrative pairs sharing an identical invariant core (T25). The red dashed curve represents the maximum likelihood exponential fit p(Delta t) = lambda e^-lambda Delta t with \hat{lambda = 1 / \mathbb{E[Delta t].

To evaluate whether the observed temporal structure arises from causal dependency, we compare the distribution against a shuffled baseline in which timestamps are randomly permuted. A KolmogorovSmirnov test yields D = 0.4917 with p < 10^-170, strongly rejecting the null hypothesis of random temporal ordering.

The inset shows the empirically estimated hazard function lambda(t), which deviates from a constant rate, indicating non-Poissonian dynamics.

These results demonstrate that counter-narrative emergence follows a structured temporal response rather than independent stochastic events.

Figure 4: Entropy Comparison (Pair vs Noise)

For each pair of narrative states, entropy is computed from the variance of their aggregated state vectors in the semantic space. We compare two conditions: (i) pairs sharing an identical invariant core (T25), and (ii) randomly sampled pairs drawn from unrelated events.

The results show a significant reduction in entropy for structurally related pairs compared to random combinations. A permutation test (1,000 iterations) yields p < 0.001, confirming that the observed entropy suppression cannot be explained by random pairing.

This effect indicates that opposing narratives derived from a shared invariant core exhibit structured interference, leading to reduced uncertainty in the combined representation. In contrast, randomly paired narratives do not exhibit this behavior, maintaining higher entropy.

These findings support the interpretation that entropy reduction is not a generic property of aggregation, but arises specifically from structurally coupled narrative states.

```
Pair     ■■■
Noise    ■■■■■■■■■
```

Figure 5: Semantic Projection Space (R)

Narrative states s \in \mathcal{S are mapped into R^3 via a projection function phi(s) that approximates the structural divergence Delta V_ij between states.

The three axes correspond to physically derived observables defined in Section 4.2: Directionality (x), Intensity (y), and Stability (z).

Distances in the projected space |phi(s*i) - phi(s*j)|*2 approximate the high-dimensional divergence Delta V*ij, enabling geometric interpretation of narrative relationships.

The projection is trained via metric learning to satisfy approximate bi-Lipschitz bounds, ensuring that relative distances are preserved within bounded distortion.

This representation provides a minimal and interpretable embedding for analyzing structural variation and clustering behavior in narrative state space.